

Chapter 13 - Physical Risk Swaps

“Physical Risk Swaps represent a straightforward extension of credit risk infrastructure rather than an exotic commodity derivative. The purpose of integrating PRS into banking is to follow Credit Default Swaps completely”. David K Kelly, MKM

Introduction

In the evolving landscape of climate finance, one critical challenge has remained unaddressed: the fundamental disconnect between long-term lending and short-term insurance coverage. This mismatch creates significant exposure for banking institutions, particularly mortgage lenders, who face escalating physical risks to their collateral assets over periods far beyond standard insurance timeframes.

“Physical Risk Swaps (PRS) represent a pioneering banking innovation designed to bridge this gap. By adapting established capital markets instruments to address climate-related physical risks—particularly flooding—PRS offers a mechanism for banking institutions to effectively transfer and manage these exposures over multi-year horizons that align with their asset portfolios.” — Johnny Mattimore, MKM

The preceding chapter traced the thirty-year evolution of Credit Default Swaps from JPMorgan’s 1994 innovation through the development of comprehensive market infrastructure. That evolution—from creative capital relief mechanism to \$62 trillion market to mature, regulated infrastructure—provides the template for Physical Risk Swaps. PRS adopts identical pricing mechanics, legal frameworks, and operational infrastructure, substituting flood events for credit events whilst maintaining complete compatibility with existing banking systems.

This chapter presents PRS as a natural extension of the CDS framework. The mathematics translate directly: hazard rates become flood probabilities, survival curves become no-flood probabilities, and the premium and protection leg calculations proceed identically. Banks’ existing credit risk infrastructure—XVA desks, pricing models, regulatory capital calculations—can process PRS without modification. The path from innovation to institutional adoption has been charted; PRS follows a proven trajectory.

Inception of Physical Risk Swaps

MKM Research Labs has been working on Sustainable Finance and Physical Risk since partnering with BNP Paribas to lead the OS-Climate “Physical Risk and Resilience” project. The concept of Physical Risk Swaps emerged from this sustained engagement with climate risk challenges facing the banking system.

In November 2023, the phrase “Physical Risk Swap” was coined by the author and Johnny Mattimore in a LinkedIn post entitled “Repricing Mortgages using Flood Insurance Risk

Transfer” as a way of describing a capital markets equivalent of parametric insurance. The natural solution was to create a new swap, building on the history of risk transfer. This eureka moment brought together several foundational principles: parametric insurance as a tradeable security; analogy with CDS product design and market development; analogy with ISDA legal definition of a CDS; and connection back to the origins of tranches, mortgage-backed securities, and collateralised mortgage obligations.

In October 2024, Johnny Mattimore and the author presented “Physical Risk Swap for Flood Risk” as a concept to the FINOS New York gathering. The driving design was to create an acceptable multi-year flood risk insurance protection in capital markets. In February 2025, Johnny Mattimore and the author presented a beta version of the Physical Risk Swap CDM alongside property, mortgage, and household insurance to the FINOS CDM Working Group.

The background to the development of Physical Risk Swaps positioned the instrument in the evolution toward risk transfer mechanisms—occupying the space where sophisticated financial engineering meets climate adaptation. This positioning reflects the conviction that climate physical risk requires not merely disclosure and measurement, but active transfer through liquid market mechanisms.

The Mortgage Lender’s Dilemma

The banking system faces a critical structural challenge in managing physical risks: a fundamental disconnect between long-term lending and short-term insurance coverage. Property owners typically borrow for 10–30 years but can only secure property insurance on annual renewal terms. This temporal mismatch creates significant exposure for banking institutions, particularly mortgage lenders, who face escalating physical risks to their collateral assets over periods far beyond standard insurance timeframes.

This disconnect represents more than just an operational inconvenience—it constitutes a growing systemic risk as continued urbanisation intensifies the financial impact of physical hazards. Banking institutions are required to reconcile the long-duration nature of their asset portfolios with the inherently short-term, annually repriced nature of traditional insurance protection. Without addressing this fundamental mismatch, lenders face increasingly uncertain risk profiles that complicate capital planning, threaten portfolio stability, and potentially restrict credit availability in vulnerable regions.

The business problem that Physical Risk Swaps address encompasses three distinct but interconnected risk layers that emerge from the insurance-lending duration mismatch:

- **Property Owners:** Face significant uncertainty about future insurability and premium costs throughout their mortgage term. Annual insurance renewals create vulnerability to sudden premium increases or, more seriously, outright insurance unavailability following major events or shifts in risk assessment. This uncertainty affects property valuations and introduces potential covenant breaches if properties become uninsurable during the loan term.

- **Mortgage Lenders:** Face direct exposure when borrowers cannot secure ongoing insurance coverage. As properties lose insurance protection, lenders bear increased risk that physical events will impair collateral values without corresponding insurance recovery. This risk compounds at the portfolio level, where geographic concentrations can simultaneously create correlated exposures across multiple properties facing insurance challenges.
- **Banking System:** Faces systemic correlation risk as weather patterns shift against further urbanisation. Entire geographic regions may experience concurrent insurance market disruptions, with insurers either raising premiums to unaffordable levels or withdrawing coverage entirely from higher-risk locations.

“This portfolio-wide concentration risk threatens synchronised collateral devaluation across significant portions of mortgage books.” — David Kelly, MKM

These risks are embedded in existing loan portfolios but remain largely unquantified in traditional risk management systems. Banking institutions must integrate physical risk assessment into their core banking architecture, measuring, monitoring, and managing these exposures with the same rigour applied to market or credit risk. They need mechanisms to translate specialised physical risk data (like flood scores from proprietary vendor models) into quantitative valuations and risk metrics for portfolio management.

For many lenders, flood risk represents the dominant physical hazard threatening collateral impairment. Yet they lack both methodologies to translate flood risk scores into banking terms and established market mechanisms for transferring this risk. This capability gap prevents active management of embedded physical risks and complicates capital allocation to climate-vulnerable regions, potentially accelerating regional economic disparities through credit availability constraints.

Parametric Insurance Foundations

The architecture of Physical Risk Swaps draws substantial inspiration from parametric insurance contracts rather than traditional catastrophe insurance. Parametric insurance triggers payouts based on objective measurements from specific instruments—such as a water level gauge on the wall of a building—without requiring an assessment of actual damages. These contracts establish direct relationships between physical measurements and banking settlements, creating transparent and rapid trigger mechanisms.

A PRS extends this concept by incorporating multiple gauges throughout a catchment area, typically managed and maintained by the relevant environmental authority, ensuring data integrity and operational reliability. This multi-gauge approach creates a comprehensive risk transfer mechanism covering broad geographic areas whilst maintaining objective, verifiable, and legally enforceable contract triggers.

By establishing tiered trigger levels corresponding to different water depths, parametric structures create granular risk segmentation that allows precise pricing and targeted protection. These established insurance constructs provide proven mechanisms for quantifying and

transferring specific segments of the risk distribution without the complications of loss adjustment processes, offering valuable precedent for capital markets approaches that require rapid and unambiguous financial settlement mechanisms. Their performance during flood events demonstrates how objective physical measurements can effectively drive banking settlements in complex environmental scenarios.

The core design of a PRS is a natural evolution of a parametric insurance contract. The only real difference is the legal wrapper, which is governed by two different regulatory frameworks. The cashflows, however, are identical. Unlike other innovations in finance, a PRS is incredibly easy to explain. Its simplicity and intuitive nature make it an elegant solution that will drive uptake.

From CDS to PRS: The Architectural Parallel

Credit Default Swaps established a well-understood pricing framework that Physical Risk Swaps adopt directly. The transition from CDS to PRS pricing requires examining the fundamental components of CDS valuation and demonstrating their direct application to physical risk.

CDS Pricing Mechanics

In a standard CDS contract, pricing involves calculating the present value of two legs: the premium leg (fixed periodic payments made by the protection buyer) and the protection leg (contingent payment made by the protection seller upon a credit event). The value of a CDS is established when the present values of these two legs are equal:

$$\text{PV Premium Leg} = \text{PV Protection Leg}$$

The premium leg payments are weighted by the probability that the reference entity survives until each payment date, whilst the protection leg is weighted by the likelihood of default. This creates a mathematical relationship:

$$\sum S \times \Delta t_i \times DF_i \times PS_i = (1 - R) \times \sum (PS_{i-1} - PS_i) \times DF_i$$

Where:

- S is the CDS spread
- Δt_i is the time interval
- DF_i is the discount factor
- PS_i is the probability of survival to time I
- R is the recovery rate.

The par spread represents the equilibrium rate where the present value of protection payments equals the present value of potential default payouts, ensuring the CDS contract has zero net value at inception.

PRS Pricing Mechanics

Physical Risk Swaps adopt this framework precisely, substituting physical events for credit events. The critical adaptation involves replacing default probability curves with physical hazard curves. For a flood PRS, the hazard curve represents the probability of a flood event of a certain magnitude occurring within a specific time frame, where meteorological data expressing “i-in-X-year” events are converted to annual exceedance probabilities.

The hazard rate $h(t)$ represents the conditional probability of the physical event occurring in a small interval Δt , given that it has not occurred before time t :

$$h(t) = \lim_{\Delta t \rightarrow 0} P(t < T \leq t + \Delta t | T > t) / \Delta t$$

For PRS, the survival probability represents the probability that the trigger event (e.g., flood level exceeding the threshold) has not occurred by time t :

$$PS(t) = e^{-\int_0^t h(s) ds}$$

The protection leg is calculated using:

$$PV \text{ Protection Leg} = (1 - R) \times \sum (PS_{i-1} - PS_i) \times DF_i$$

Where:

R represents the “recovery value” in the physical risk context—the residual value after the physical event occurs.

The spread for a PRS represents the cost of protection and is calculated identically to a CDS spread:

$$S = [(1 - R) \times \sum (PS_{i-1} - PS_i) \times DF_i] / [\sum \Delta t_i \times DF_i \times PS_i]$$

According to market conventions for standardised contracts, this spread is converted to an upfront payment plus a fixed coupon, mirroring the Big Bang Protocol conventions established for CDS.

Service Lines: CDS to PRS Mapping

The commercial opportunity for PRS mirrors that proven for CDS. The service lines that emerged around credit derivatives—from IHS Markit/S&P—translate directly to physical risk:

Service Category	CDS (IHS Markit/S&P)	PRS (MKM)
Elementary Services	CDS Pricing Data	PRS Pricing Data (Gauge/Bench)
	Reference Entity Data (RED)	Physical Asset Entity including C
	Credit Indices Administration	PR Indices & Migration Matrices
Intermediate Services	Sector & Rating Curves	Regional Catchment (Polygon) I
	Enhanced Sector/Rating Curves	Enhanced Regional Catchment C
	Liquidity & Market Metrics	Liquidity & Market Metrics
	Credit Event/Corporate Action	PR Event/Adaptation Services
	Historical Archives & Surveillan	Historical Archives & Surveillan
Advanced Services	Loan & Securitised Credit Data	Structured Credit PR Enhancem
	Entity Matching & Cross-Refer	Gauge Matching & Cross-Refer
	Portfolio Valuations	Portfolio Valuations
	Trade Processing (MarkitSERV	Trade Processing & Lifecycle

	Research Signals & Factor Libr	Research Signals & Factor Libr
	Custom Curves & Bespoke Proc	Custom Curves & Bespoke Proc
	Regulatory & Compliance Supp	Regulatory & Compliance Supp
Professional Services	Professional Services & Advisory	Professional Services & Advisory
	Platform & Data Integration	Platform & Data Integration

This parallel demonstrates that PRS maps identically to CDS for market instrument standards (trading OTC to documentation via ISDA), market infrastructure (give-up, clearing, compression), and bank risk architecture (models, systems, data, capital calculations and relief). The most significant commercial opportunity lies in the lifecycle of indices, precisely as proven for CDS.

From Return Periods to Hazard Curves

Traditional flood risk assessment employs return periods: a “1-in-100-year flood” indicates an event with 1% annual exceedance probability. Whilst intuitive for communication, return periods require translation into the continuous hazard rate framework that credit risk systems expect.

The Fundamental Relationship

The mathematical relationship between return period and exceedance probability provides the starting point. For a return period τ (in years), the probability of at least one exceedance event in time period t follows:

$$P(X \geq 1) = 1 - e^{-(t/\tau)}$$

For annual assessment ($t = 1$ year), a 100-year return period yields approximately 1% annual exceedance probability. For a trigger level with annual exceedance probability p , the constant hazard rate λ satisfies:

$$1 - e^{(-\lambda)} = p$$

Solving yields $\lambda = -\ln(1-p)$. For small probabilities ($p < 10\%$), the approximation $\lambda \approx p$ holds with minimal error, enabling straightforward translation from return period data to hazard rate parameters.

Gauge-Specific Hazard Curve Construction

Each monitoring gauge possesses unique flood characteristics captured through historical data and climate projections. A gauge might exhibit the following return period profile:

Return Period	Flood Level	Annual Exceedance Probability
10 years	3.5 metres	10.0%
25 years	4.0 metres	4.0%
50 years	4.2 metres	2.0%
100 years	4.8 metres	1.0%
200 years	5.2 metres	0.5%
500 years	5.5 metres	0.2%

These discrete data points fit a continuous curve representing the relationship between flood levels and exceedance probabilities. The trigger level specified in the PRS contract determines which point on this curve applies. A contract triggered at 4.8 metres references the 1% annual exceedance probability; a contract triggered at 5.5 metres references the 0.2% annual exceedance probability.

The probability space for pricing is built by converting traditional flood return periods into annual exceedance probabilities, developing a continuous hazard rate function based on these probabilities, and incorporating weather projections to adjust forward-looking probabilities. Each monitoring gauge has its hazard curve based on historical data and climate projections.

Event Definition and Triggering Mechanism

Credit Default Swaps and Physical Risk Swaps share structural similarities but differ fundamentally in their triggering mechanisms and applications. Both instruments allow for customisable risk transfer, but PRS leverage measurable physical parameters—such as flood depths—to define payout conditions, offering granular flexibility akin to credit risk tranching in CDS markets.

CDS Triggering

CDS contracts are structured to cover credit events across a spectrum of credit grades (AA to B), with payouts triggered by predefined conditions like bankruptcy, failure to pay, or restructuring. Key features include dynamic pricing (CDS spreads adjust intraday based on market perceptions of creditworthiness, enabling trading at intermediate risk levels before a default event) and recovery rates (post-default payouts account for the residual value of debt, typically determined via cash settlement).

PRS Triggering

PRS contracts can define payout triggers for specific flood depths (e.g., 0.5m–1.0m) or catastrophic levels (e.g., >2.0m), analogous to tranching CDS. Crucially, flood events are determined by remote sensors or gauge data, eliminating subjective judgements required in credit events like restructuring. This objectivity represents a significant advantage: whilst CDS requires Determination Committees to adjudicate whether restructuring constitutes a credit event, PRS triggers are binary and verifiable—the gauge either exceeded the threshold or it did not.

A flood PRS typically references specific water level gauges in a catchment area. Many contracts reference multiple gauges, requiring correlation modelling between different gauge locations, settlement conditions based on one or more gauges exceeding thresholds, and weighted payout structures based on the severity and location of flooding. The correlation modelling typically employs copula functions to represent joint probability distributions between gauges, accounting for the fact that flood events often affect multiple locations simultaneously.

Product Tiers: Index, Portfolio, and PRO

The PRS platform offers three product tiers addressing different risk management needs and sophistication levels, mirroring the evolution of CDS from single-name contracts to index products to tranching structures.

Index PRS

The entry-level product provides liquid, standardised protection. Single-gauge triggers (e.g., Thames at Teddington at 5.5m) produce all-or-nothing payouts. At 55bp baseline pricing for a three-year term, Index PRS offers the lowest cost and simplest structure. One-day execution makes this suitable for rapid hedging requirements. The trade-off is basis risk: portfolio-specific characteristics remain unhedged.

Characteristic	Index PRS
Trigger	Single gauge (Thames @ Teddington 5.5m)
Payout	All-or-nothing (£500M or £0)
Price	55bp (baseline)

Execution	1 day
Pros	Liquid market, simple, easy to understand
Cons	High basis risk, binary payout, no customisation
Typical Buyer	Retail investors, quick hedges

Portfolio PRS

The institutional tier provides customised protection matched to specific portfolio characteristics. Multi-gauge weighted triggers (e.g., 60% Thames, 40% Richmond) combined with asset location data eliminate basis risk. At 63bp (8bp above index), the additional cost buys precision. Execution requires two to three weeks for customisation and documentation. Institutional funds and REIT portfolios with defined physical risk exposure benefit most from this tier.

Characteristic	Portfolio PRS
Trigger	Multi-gauge weighted (60/40 Thames/Richmond) + asset locations
Payout	All-or-nothing but portfolio-specific
Price	63bp (+8bp basis vs index)
Execution	2-3 weeks
Pros	Zero basis risk, customised to assets
Cons	More expensive, less liquid, longer execution
Typical Buyer	Institutional funds, REIT portfolios

PRO Tranches (Portfolio Risk Obligation)

The sophisticated tier introduces CDO-style structuring for flood risk. Rather than all-or-nothing payouts, PRO products split exposure into tranches with graduated triggers. This structure enables risk layering: sophisticated hedgers can purchase only the tranches matching their risk appetite, whilst yield-seeking investors can access the equity tranche's attractive returns.

Tranche	Flood Depth	Notional	Yield	Risk Profile
Equity	0-0.2m	£50M	180bp	First loss, high yield
Mezzanine	0.2-1.0m	£150M	85bp	Subordinated
Senior	1.0m+	£300M	35bp	Protected, low yield

The blended cost of 64.5bp across all tranches provides optionality at minimal premium to Portfolio PRS. Each tranche absorbs losses in order: equity absorbs first (0-0.2m flood depth), mezzanine absorbs next (0.2-1.0m), and senior absorbs only catastrophic losses (>1.0m). This mirrors CDO waterfall structures familiar from credit markets.

Portfolio PRS Basis Decomposition

The Portfolio PRS pricing differs from Index PRS due to “basis”—the additional spread reflecting the distance from assets to reference gauge, elevation differences (topography), hydrological correlations (water flow patterns), and model uncertainty and composition effects. Basis is the key innovation that enables precise, customised flood hedging.

Component	Impact	Typical Range	Example (Thames)
Base Index PRS	Baseline	N/A	55bp (3Y)
Model Uncertainty	Spread on model inputs	±2bp	+2bp
Distance to Gauge	Farther = higher risk	+1bp to +6bp	+3bp
Terrain Variation	Flood plain exposure	-1bp to +3bp	+2bp
Elevation Benefit	Higher assets safer	-2bp to 0bp	-1bp
Asset Composition	Diversity effects	-1bp to +2bp	+2bp
Total Basis	Sum of components	±3bp to ±8bp	+8bp
Portfolio PRS	Final price	Index ± Basis	63bp

The basis waterfall provides transparency: every basis point differential receives explicit justification through quantified risk factors. This transparency distinguishes the platform from opaque insurance pricing and supports informed hedging decisions.

The Basis Risk Decision

Consider the Richmond REIT portfolio: £500M across 120 assets. Index PRS at 55bp provides protection triggered by the Thames at Teddington reaching 5.5 metres. However, portfolio assets average 2.3km from the gauge with +15m average elevation. The index hedge covers only 64% of actual portfolio risk (beta 0.64), leaving scenarios where assets flood but the gauge trigger is not reached—basis losses that could reach several million pounds.

Portfolio PRS at 63bp eliminates this basis risk by triggering on actual portfolio flood impact rather than index gauge levels. The additional 8bp annual cost translates to £1.2M over a three-year term for a £500M portfolio—well justified against potential multi-million basis losses. The extra £1.2M premium is well worth eliminating multi-million basis risk exposure. The REIT avoids scenarios where assets flood but the gauge trigger is not hit, leaving them unhedged.

Implementation Framework

The journey from theoretical pricing models to a functioning Physical Risk Swap market requires careful orchestration across regulatory, technological, and banking dimensions. Rather than developing entirely new market infrastructure, the PRS ecosystem leverages existing frameworks whilst introducing targeted innovations to address the unique characteristics of physical risk.

The transformative innovation of PRS lies in extending established risk transfer concepts into banking and capital markets infrastructure. Whilst insurance mechanisms have long addressed physical risks, they remain largely separate from banking capital frameworks and lack the standardisation, liquidity, and regulatory integration necessary for effective banking risk management. PRS bridge this gap by aligning with existing data sets from environmental monitoring and catastrophe modelling whilst structuring the banking instrument according to ISDA standards—the same legal and operational framework that underpins hundreds of trillions in interest rate and credit derivatives.

“Most significantly, PRS development includes regulatory capital methodologies aligned with Basel standards, enabling banks to receive appropriate capital relief for transferred physical risks and creating powerful incentives for market adoption.” — David Kelly, MKM

Leveraging Existing Monitoring Infrastructure

Implementing PRS benefits from substantial existing infrastructure that can be leveraged without requiring extensive new monitoring systems. For example, the Thames River Basin in the UK represents an ideal development environment with its highly populated, economically critical areas and persistent flood risks affecting assets from residential to industrial properties.

The UK already maintains a robust flood monitoring system managed by government agencies. A network of monitoring stations along the Thames River provides real-time water level data at regular intervals. This existing infrastructure offers a solid foundation for flood risk assessment models and creates reliable triggering mechanisms for banking contracts. Rather than building

new monitoring capabilities, PRS implementation can focus on integrating and standardising these data streams for banking applications.

Strategic Partnerships

Strategic partnerships with leading experts in flood modelling allow the adaptation of terrain mapping, flood simulators, and event set development to meet the specific needs of mortgage lending risk assessment. These collaborations enable access to decades of accumulated expertise in hydrological modelling without requiring banking institutions to develop specialised climate science capabilities internally.

The computational demands of PRS valuation require partnerships providing access to high-performance computing capabilities. Whilst gauge-level monitoring data is available in real-time, the complex task of portfolio revaluation requires substantial computing resources, particularly for stress testing across multiple climate scenarios. Cloud-based computing platforms with dedicated physical risk modelling capabilities allow market participants to access analytics without individual infrastructure investments.

For effective implementation, governance frameworks must include cross-disciplinary representation where meteorologists, hydrologists, atmospheric and weather scientists work alongside derivatives structurers and trading strategists to refine risk assessment methodologies. Insurance underwriters with decades of catastrophe experience advise on threshold calibration and recovery value estimation. This cross-disciplinary integration accelerates product development whilst strengthening model validation through diverse perspectives.

ISDA Master Agreement and Market Practices

The cornerstone of derivatives market standardisation for decades—the ISDA Master Agreement—provides the contractual foundation for Physical Risk Swaps. This established legal framework reduces documentation uncertainty and creates immediate familiarity for market participants who have executed trillions in interest rate and credit derivative trades under the same structure. The agreement’s modular design accommodates new definitions and supplements for physical risk events whilst maintaining consistency with existing market conventions for trade confirmation, settlement, and dispute resolution.

FINOS Common Domain Model

This legal standardisation is complemented by the FINOS Common Domain Model (CDM), which offers critical data standardisation across the trade lifecycle. The CDM creates a unified representation of trade events, product definitions, and process models, enabling seamless interoperability between counterparties, clearing houses, and regulatory reporting systems.

By incorporating physical risk parameters and gauge-level monitoring data into this existing framework, PRS transactions benefit from day-one compatibility with established market infrastructure. The CDM implementation ensures that gauge readings, threshold triggers, and

settlement calculations maintain consistency across diverse market participants and technology platforms.

Digital Regulatory Reporting

Most significant for market adoption, Physical Risk Swaps integrate directly with the FINOS Digital Regulatory Reporting (DRR) framework from inception. This pre-established compliance approach eliminates the traditional regulatory uncertainty often accompanying banking innovation. Because PRS instruments are designed to be DRR-compliant from their first transaction, they avoid the regulatory ambiguity and implementation lags that have historically slowed the adoption of new derivative products. Banks can confidently deploy these instruments with clear regulatory treatment, focusing implementation efforts on risk modelling rather than compliance infrastructure.

Developing Market Liquidity

Developing a liquid market for Physical Risk Swaps represents a critical innovation in risk management for physical assets. Following the successful evolution of the Credit Default Swap market, we can establish a roadmap for creating a robust, efficient marketplace for transferring and managing physical risk exposures.

Learning from the CDS Market Evolution

The Credit Default Swap market demonstrated how standardisation, institutional participation, and regulatory frameworks can create a functioning risk transfer mechanism. From its inception in the early 1990s to its peak of \$62 trillion in notional value by 2007, the CDS market grew on the back of infrastructure that PRS can now inherit. Whilst the market subsequently contracted to approximately \$9 trillion, the core infrastructure and liquidity mechanisms remained intact, particularly for index products.

The core design of a PRS is a natural evolution of a parametric insurance contract. The only real difference is the legal wrapper, which is governed by two different regulatory frameworks. The cashflows, however, are identical. Unlike other innovations in finance, a PRS is incredibly easy to explain. Its simplicity and intuitive nature make it an elegant solution that will drive uptake.

Liquidity Development Trajectory

Based on the CDS market experience, we can anticipate the following liquidity development pattern for PRS:

Initial Phase: Concentrated bilateral trading among major dealers using standardised documentation.

Index Development: Creation of diversified indices covering major risk categories and geographies (Thames, Mississippi, Rhine, Han River).

Buy-Side Adoption: Gradual expansion to asset managers, pension funds, and other institutional investors.

Market Bifurcation: Emergence of highly liquid index products alongside more specialised single-name PRS.

Electronic Execution: Migration from voice trading to electronic platforms as volume increases.

Market Maturity: Development of secondary market trading, curve trading, and basis trading strategies.

Central Clearing Advantages

Early central clearing provides significant advantages for market development, as demonstrated by the CDS market's post-crisis evolution. Central clearing delivers counterparty risk reduction through a central counterparty, netting efficiency improving capital efficiency through multilateral netting, standardised margining creating predictable and efficient collateral requirements, and position portability facilitating the transfer of positions between counterparties.

The involvement of major banking institutions is critical for initial liquidity, requiring a coordinated launch and ongoing commitment to make markets by major banks that create an immediate two-way market.

Market Impact: Beyond Risk Transfer

Whilst mortgage lenders are the primary beneficiaries, Physical Risk Swaps have broader implications for property markets, climate resilience, and banking stability.

The introduction of PRS creates significant positive externalities beyond direct risk transfer benefits. By reducing long-term risk exposure, lenders may be more willing to maintain mortgage availability in flood-prone areas, improving property market liquidity in regions that might otherwise face credit contraction. This sustained credit provision supports property values and enables ongoing investment in areas requiring adaptation rather than abandonment.

PRS pricing provides valuable market-based information about perceived flood risks, creating transparent signals that inform urban planning, infrastructure investment, and adaptation priorities. Regional pricing differentials will highlight areas where flood mitigation infrastructure could deliver the most significant economic benefits as the market develops, potentially influencing public investment decisions and encouraging preventative measures rather than post-disaster recovery.

The availability of PRS may incentivise community-wide resilience improvements by creating banking rewards for risk reduction. As lenders better manage their risks, they may offer more favourable terms to borrowers implementing flood resilience measures, creating economic incentives for adaptation investments.

The broader banking system benefits from improved risk transparency and reduced correlation of flood-related losses. By distributing concentrated physical risks more widely through capital markets, PRS minimises the potential for synchronised institutional stress during significant weather events. This distribution mechanism supports banking stability objectives by preventing the concentration of weather risks in institutions with limited capacity to bear them.

Market Size and Commercial Opportunity

The potential scale of the Physical Risk Swap market is substantial, with estimates suggesting it could develop into a multi-trillion dollar notional market as significant capital markets firms integrate these instruments into their offerings. This projected growth follows patterns observed in other successful derivatives markets.

For comparison, CDS currently represents approximately \$10 trillion in gross notional, whilst interest rate derivatives are in the hundreds of trillions. The annual notional value of PRS transactions may realistically grow into the trillions of USD as the market matures.

First-mover advantage creates network effects and establishes market convention around pricing methodologies. This would position early entrants as physical risk/climate risk market infrastructure—analogue to interest rate and credit derivative utilities—capturing transaction fees, index licensing revenue, and calculation agent mandates.

Conclusion

Physical Risk Swaps represent not merely a technical banking innovation but a critical step toward a banking system that properly accounts for and manages physical risks—ultimately supporting both economic stability and climate adaptation in the decades ahead.

The successful implementation of PRS requires collaboration between atmospheric and weather scientists, hazard assessment experts, banking engineers, and market participants to develop robust pricing models that accurately capture the complex dynamics of physical risks whilst maintaining the standardisation and liquidity necessary for market adoption. With the foundation of ISDA documentation and the FINOS CDM, the market can focus on the unique aspects of physical hazard risk modelling rather than reinventing established market infrastructure.

The following chapter presents the operational platform that brings these concepts to life: the PRS Pricing Platform with its real-time gauge monitoring, hazard curve construction, portfolio basis calculation, and historical validation capabilities.